

Physics & Core Algorithms

This document outlines the governing equations, thermodynamic models, and numerical solvers utilized by the Refrigerator Volume & Thermal Cycle Calculator.

1. Geometric Mathematics

1.1. Stepped Floor Cross-Section (Shoelace Formula)

To accurately calculate the internal volume of a compartment containing the compressor step, the system constructs a 2D bounding polygon. The area of this irregular polygon is evaluated using the Shoelace formula (Gauss's area formula):

$$A = \frac{1}{2} \left| \sum_{i=1}^{n-1} (x_i y_{i+1} - x_{i+1} y_i) + (x_n y_1 - x_1 y_n) \right|$$

Where (x_i, y_i) are the vertices of the compartment's inner profile, compensating for the angled compressor step.

2. Heat Load Modeling

Heat loads are calculated by summing the thermal conduction through the cabinet walls, doors, and internal dividers.

2.1. Thermal Conductivity of Polyurethane Foam

The thermal conductivity λ of the rigid polyurethane foam insulation varies with the average wall temperature $T_{avg} = \frac{T_{in} + T_{out}}{2}$. The base empirical formula is adjusted by a conversion factor (1.16279) to map $kcal/(h \cdot m \cdot K)$ to $W/(m \cdot K)$:

$$\lambda = (0.0165 + 0.00011 \cdot (T_{avg} - 25)) \cdot 1.16279$$

2.2. Overall Heat Transfer Coefficient (U -value)

Heat transmission through the cabinet incorporates surface film coefficients ($h_{out} \approx 6.977 W/m^2 K$, $h_{in} \approx 11.628 W/m^2 K$).

The reciprocal sum determines

the effective transmission coefficient U :

$$U_{exterior} = \frac{1}{\frac{1}{h_{out}} + \frac{1}{h_{in}} + \frac{t}{\lambda}}$$

$$U_{interior} = \frac{1}{\frac{1}{h_{in}} + \frac{1}{h_{in}} + \frac{t}{\lambda}}$$

Where t is the insulation thickness in meters. The heat load for any given face is thus $Q = U \cdot A \cdot \Delta T$.

3. Evaporator & Airflow Physics

3.1. Air Velocity

Airflow speed over the evaporator matrix is derived from the volumetric flow rate of the fan:

$$v = \frac{\dot{V}_{fan}}{A_{face} \cdot 3600}$$

Where $\dot{V}_{fan}$ is in m^3/h , A_{face} is the frontal cross-sectional area of the evaporator in m^2 , and v is in m/s .

3.2. Convective Heat Transfer Coefficient

The convective heat transfer coefficient α of the evaporator fins is modeled empirically based on air velocity:

$$\alpha = 12.93 \cdot v^{0.415} \cdot 1.16279 \quad [W/(m^2 \cdot K)]$$

3.3. Log Mean Temperature Difference (LMTD)

The heat transfer capacity of the evaporator relies on the logarithmic mean temperature difference between the mixed return air T_1 , the supply air T_2 , and the uniform evaporating

refrigerant temperature T_E :

$$LMTD = \frac{(T_1 - T_E) - (T_2 - T_E)}{\ln \left(\frac{T_1 - T_E}{T_2 - T_E} \right)}$$

Total evaporator capacity is:

$$Q_{EV} = \alpha \cdot A_{total} \cdot LMTD$$

4. Compressor & Refrigerant Modeling

4.1. Constant-Speed Compressor Polynomials

Compressor performance is modeled using standard AHRI-style polynomials derived via Ordinary Least Squares (Gauss-Jordan elimination with partial pivoting) from empirical test data.

Volumetric Efficiency (η_v):

$$\eta_v = A + B \left(\frac{P_c}{P_e} \right) + C \cdot P_c$$

Input Power (W):

$$W = AW + BW \cdot T_E + CW \cdot T_C + DW \cdot T_C \cdot T_E + EW \cdot T_E^2$$

4.2. Inverter Compressor Models

Variable-speed compressors use a Ridge Regression model. The system evaluates multiple features (linear and quadratic) based on normalized RPM (n):

$$W, Q = \beta_0 + \beta_1 n + \beta_2 (n \cdot T_E) + \beta_3 (n \cdot T_C) + \beta_4 (n \cdot T_C \cdot T_E) + \beta_5 (n \cdot T_E^2)$$

The system automatically cross-validates log-transformed ($\ln(n)$) and piecewise RPM models, selecting the configuration with the lowest RMSE.

4.3. Mass Flow Calculation

Theoretical mass flow is derived from the compressor's swept volume V_c (cm^3), speed N (RPM), and the specific volume of the suction gas v_{gas} (m^3/kg):

$$\dot{m}_{theoretical} = \frac{V_c \cdot N \cdot 60}{10^6 \cdot v_{gas}}$$

Actual mass flow is $\dot{m}_{actual} = \eta_v \cdot \dot{m}_{theoretical}$.

5. Condenser Heat Rejection

Condenser heat rejection capacity $Q_{C,out}$ evaluates the side and back skin condensers. The heat transfer multiplier K is defined empirically by the pipe pitch p (in mm):

$$K = 1.0738 - 0.004152 \cdot p + 0.00000482 \cdot p^2$$

Total side heat rejection:

$$Q_{side} = K_{side} \cdot A_{side} \cdot (T_C - T_0) \cdot 1.16279$$

6. Numerical Solver Architecture

The thermodynamic state is resolved using a nested iterative solver to balance the thermal loads against the mechanical capacity.

6.1. The Inner Loop (Newton-Raphson)

Resolves the air-side and running ratio parameters for a *guessed* condensing temperature T_C .

- **Variables:** Evaporator exit temperature T_2 and Compressor Running Ratio PR (or RPM for inverters).
- **Objective Functions:**
 1. $F_1 = Q_F - \dot{m}_{air,F} \cdot C_{p,air} \cdot (T_F - T_3) \cdot PR = 0$ (Freezer load balance)
 2. $F_2 = Q_{total} - Q_{comp} \cdot PR = 0$ (Total system load balance)
- **Method:** 2x2 Jacobian matrix evaluated via finite differences with backtracking line search to guarantee sufficient decrease.

6.2. The Evaporator Temperature (T_E) Loop

For a given mixed air inlet T_1 and supply T_2 , the exact required T_E is found using **Brent's Method**. It solves the non-linear relationship where the actual $LMTD$ equals the required $LMTD$ calculated from $Q_{total} / (\alpha \cdot A)$.

6.3. The Outer Loop (Secant Method)

Resolves the system condensing temperature T_C .

- **Objective Function:** Balances the heat rejected by the condenser surface ($Q_{C,out}$) against the heat pumped into the condenser by the compressor ($Q_{C,in} = \dot{m}_{actual} \cdot \Delta h$):

$$F_3(T_C) = Q_{C,out} - Q_{C,in} = 0$$

- **Method:** Secant root-finding method using forward/backward perturbations ($\Delta T_C = 0.001K$) to approximate the derivative $\frac{dF_3}{dT_C}$. Iterates until $|F_3| < 0.001$ W.